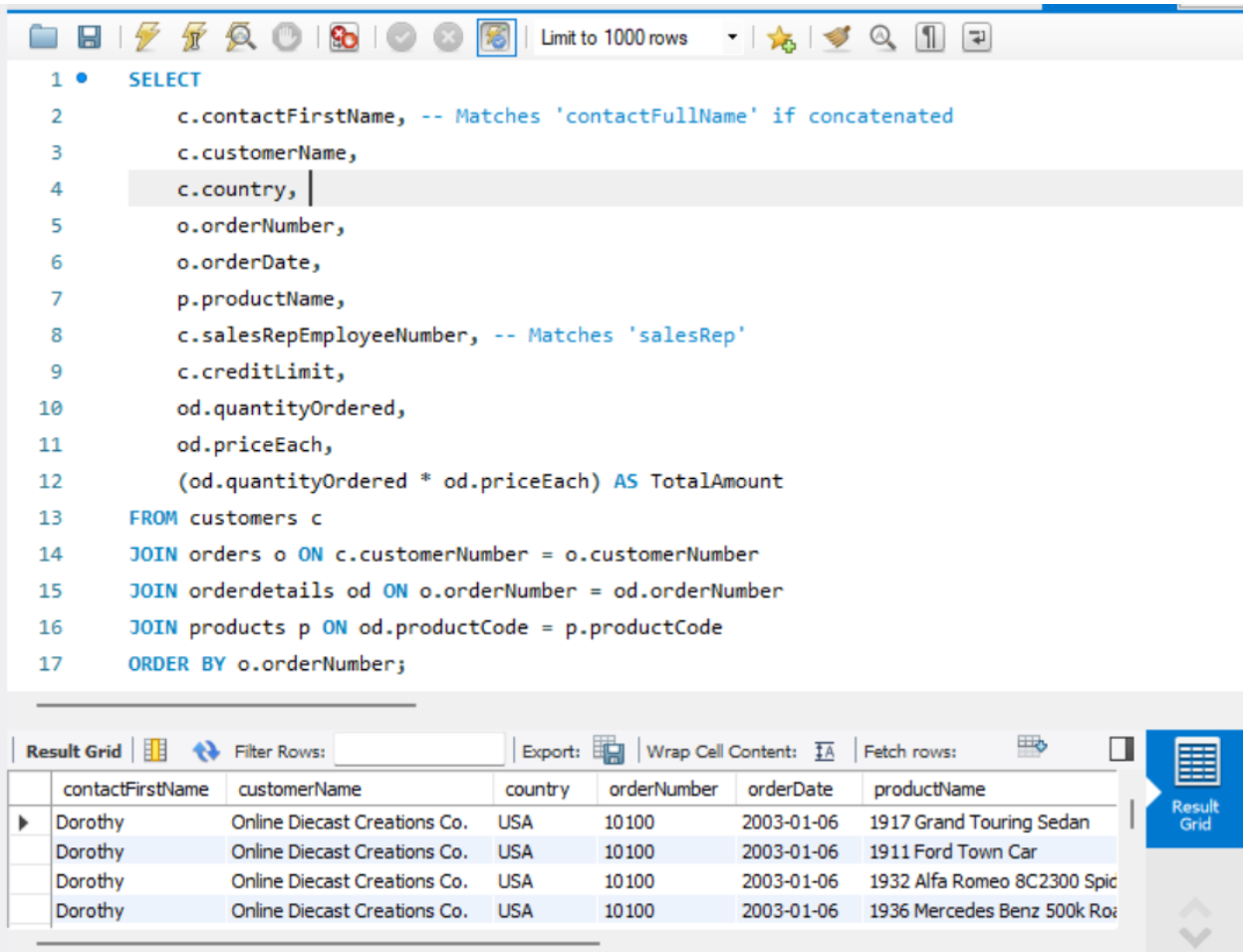
EXPLORATORY DATA ANALYSIS

USING SQL TO CLEAN, TRANSFORM AND ANALYSE

This project focuses on a company dealing in classic, vintage, and collector automobiles. The dataset, MintClassics, contains information on sales, products, services, and operational processes.

The first step is to explore and extract our database which is done by uploading the MintClassics database file into MySQL Workbench. The data was further explored and exported to Excel to identify clean, remove duplicates and mismatches and extract the relevant information from the dataset. The selected data was then imported into a CSV file for further processing.

This section includes a screenshot of the SQL command used to extract the required data.

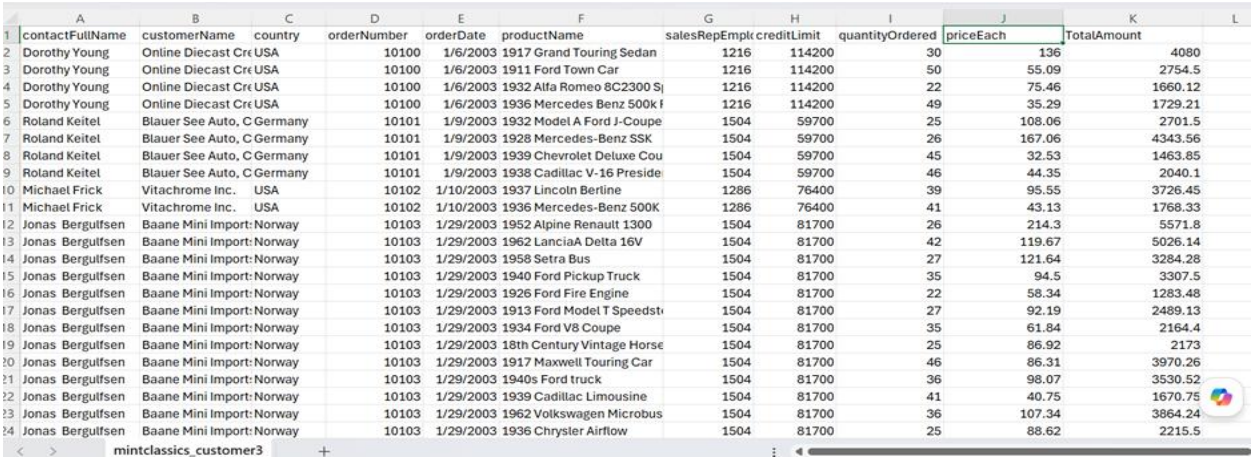


```sql
SELECT
    c.contactFirstName, -- Matches 'contactFullName' if concatenated
    c.customerName,
    c.country,
    o.orderNumber,
    o.orderDate,
    p.productName,
    c.salesRepEmployeeNumber, -- Matches 'salesRep'
    c.creditLimit,
    od.quantityOrdered,
    od.priceEach,
    (od.quantityOrdered * od.priceEach) AS TotalAmount
FROM customers c
JOIN orders o ON c.customerNumber = o.customerNumber
JOIN orderdetails od ON o.orderNumber = od.orderNumber
JOIN products p ON od.productCode = p.productCode
ORDER BY o.orderNumber;
```

Result Grid

contactFirstName	customerName	country	orderNumber	orderDate	productName
Dorothy	Online Diecast Creations Co.	USA	10100	2003-01-06	1917 Grand Touring Sedan
Dorothy	Online Diecast Creations Co.	USA	10100	2003-01-06	1911 Ford Town Car
Dorothy	Online Diecast Creations Co.	USA	10100	2003-01-06	1932 Alfa Romeo 8C2300 Spic
Dorothy	Online Diecast Creations Co.	USA	10100	2003-01-06	1936 Mercedes Benz 500k Roa

A MySQL database was then created, and the cleaned data was imported into MySQL Workbench. Although the dataset was already cleaned in Excel, SQL was still used for further data cleaning and transformation as a form of the ETL (Extract, Transform, Load) process before analysis.



contactFullName	customerName	country	orderNumber	orderDate	productName	salesRepEmpl	creditLimit	quantityOrdered	priceEach	TotalAmount
Dorothy Young	Online Diecast Cre	USA	10100	1/6/2003	1917 Grand Touring Sedan	1216	114200	30	136	4080
Dorothy Young	Online Diecast Cre	USA	10100	1/6/2003	1911 Ford Town Car	1216	114200	50	55.09	2754.5
Dorothy Young	Online Diecast Cre	USA	10100	1/6/2003	1932 Alfa Romeo 8C2300 S	1216	114200	22	75.46	1660.12
Dorothy Young	Online Diecast Cre	USA	10100	1/6/2003	1936 Mercedes Benz 500k I	1216	114200	49	35.29	1729.21
Roland Keitel	Blauer See Auto, C	Germany	10101	1/9/2003	1932 Model A Ford J-Coupe	1504	59700	25	108.06	2701.5
Roland Keitel	Blauer See Auto, C	Germany	10101	1/9/2003	1928 Mercedes-Benz SSK	1504	59700	26	167.06	4343.56
Roland Keitel	Blauer See Auto, C	Germany	10101	1/9/2003	1939 Chevrolet Deluxe Cou	1504	59700	45	32.53	1463.85
Roland Keitel	Blauer See Auto, C	Germany	10101	1/9/2003	1938 Cadillac V-16 Preside	1504	59700	46	44.35	2040.1
Michael Frick	Vitachrome Inc.	USA	10102	1/10/2003	1937 Lincoln Berline	1286	76400	39	95.55	3726.45
Michael Frick	Vitachrome Inc.	USA	10102	1/10/2003	1936 Mercedes-Benz 500K	1286	76400	41	43.13	1768.33
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1952 Alpine Renault 1300	1504	81700	26	214.3	5571.8
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1962 LanciaA Delta 16V	1504	81700	42	119.67	5026.14
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1958 Setra Bus	1504	81700	27	121.64	3284.28
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1940 Ford Pickup Truck	1504	81700	35	94.5	3307.5
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1926 Ford Fire Engine	1504	81700	22	58.34	1283.48
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1913 Ford Model T Speedst	1504	81700	27	92.19	2489.13
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1934 Ford V8 Coupe	1504	81700	35	61.84	2164.4
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	18th Century Vintage Horse	1504	81700	25	86.92	2173
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1917 Maxwell Touring Car	1504	81700	46	86.31	3970.26
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1940s Ford truck	1504	81700	36	98.07	3530.52
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1939 Cadillac Limousine	1504	81700	41	40.75	1670.75
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1962 Volkswagen Microbus	1504	81700	36	107.34	3864.24
Jonas Bergulfsen	Baane Mini Import:	Norway	10103	1/29/2003	1936 Chrysler Airflow	1504	81700	25	88.62	2215.5

mintclassics_customer3

Here are the primary types of data cleaning and transformation tasks typically performed using SQL in this project:

Typical Transformation Workflow for MintClassics:

Type used	SQL Operation Command Used	Purpose
Cleaning	**WHERE, COALESCE, UPPER**	**Removes noise and standardizes inputs.**
Transformation	**SUM, AVG, CASE**	**Creates KPIs like "Revenue" or "Inventory Turnover."**
Structuring	**JOIN, UNION, WITH (CTEs)**	**Combines disparate tables into one cohesive report.**

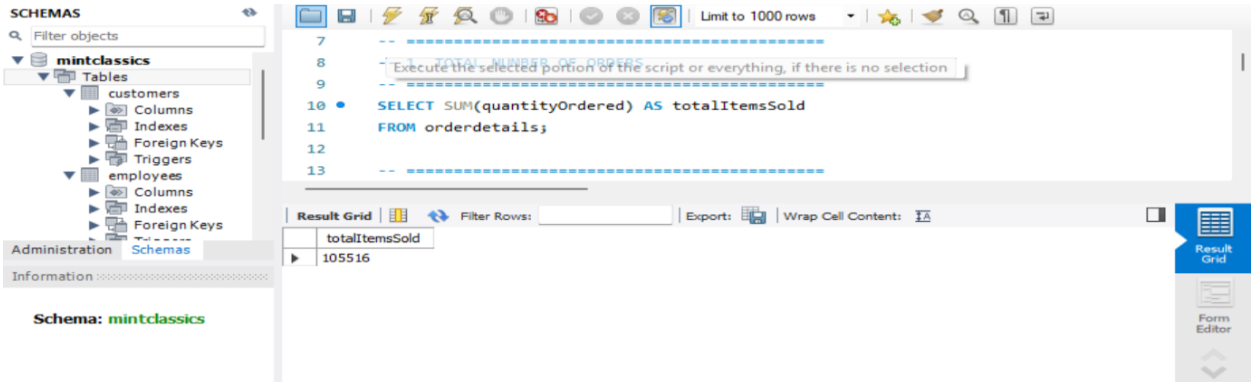
Example SQL does not understand this date format 1/6/2003 therefore transformed into Year- Month-Date, thou day was excluded in the queries used in creating in Analysis 7.

You can check it for better understanding

salesYear	salesMonth	totalRevenue
2003	January	116692.77
2003	February	128403.64
2003	March	160517.14
2003	April	185848.59
2003	May	179435.55
2003	June	150470.77
2003	July	201940.36
2003	August	178257.11
2003	September	236697.85
2003	October	514336.21
		292385.21

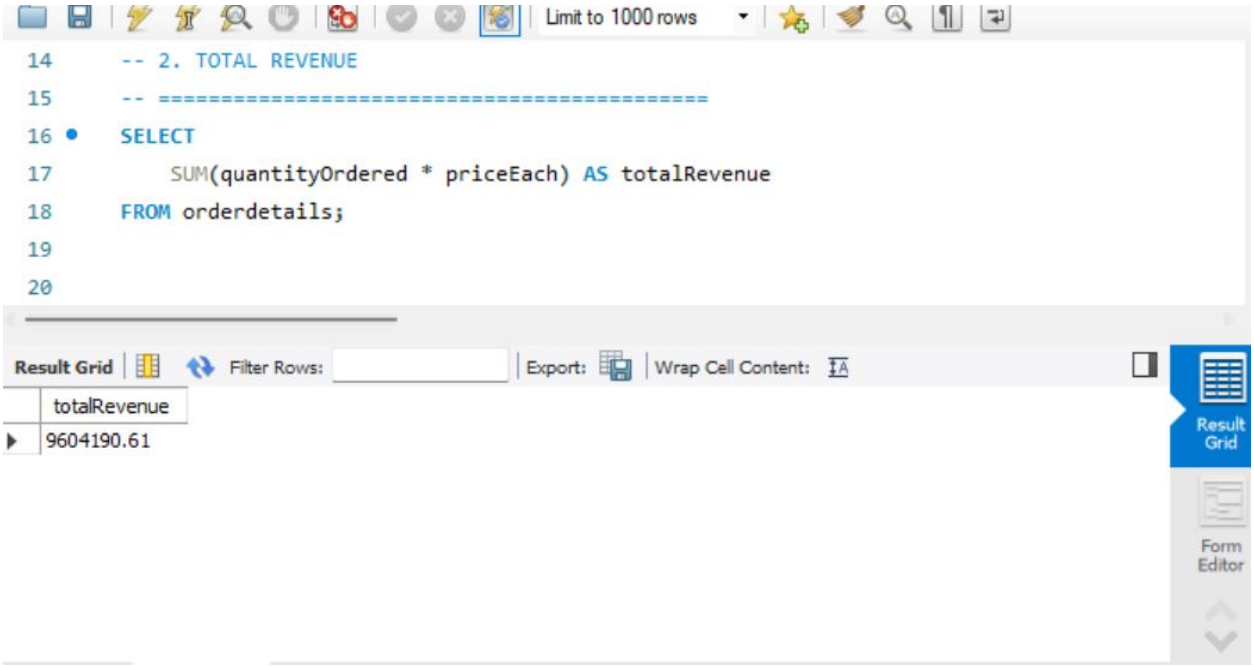
ANALYSIS 1: TOTAL PRODUCTS ORDERED

Using these SQL queries with SUM, total Products ordered were calculated



ANALYSIS 2: TOTAL REVENUE

Calculate the total revenue using an SQL query.



ANALYSIS 3: COUNTRIES WITH HIGHEST SALES

With SQL queries using LEFT JOIN, GROUPBY, Countries with highest Sales limited to 10.

This analysis is based on aggregated sales data, ensuring accurate comparison across countries."

```
22    -- ============================================
23    -- 4. COUNTRY SALES PERFORMANCE
24    -- ============================================
25    SELECT
26        c.country,
27        SUM(od.quantityOrdered * od.priceEach) AS totalRevenue
28    FROM customers c
29    JOIN orders o ON c.customerNumber = o.customerNumber
30    JOIN orderdetails od ON o.orderNumber = od.orderNumber
31    GROUP BY c.country
32    LIMIT 10;
33
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

country	totalRevenue
France	1007374.02
USA	3273280.05
Australia	562582.59
Norway	270846.30
Germany	196470.99
Spain	1099389.09
Sweden	187638.35
Denmark	218994.92

Result 78 | Result 79 | Result 80 | Result 81 | Result 82 | Result 83 | Result 84 | Result 85 | Read Only

Result Grid | Form Editor

ANALYSIS 4: STOCK IN WAREHOUSE

Understanding Products quantity in Stock across warehouse

The SQL commands used are structuring, **JOIN, FROM, and ORDERBY.**

This insight supports better demand planning, inventory balancing, and cost control strategies especially in the case of warehouse utilization

```
36    -- 5. STOCK IN WAREHOUSE
37    -- ========================================
38    SELECT
39        w.warehouseName,
40        p.productName,
41        p.quantityInStock
42    FROM products p
43    JOIN warehouses w
44        ON p.warehouseCode = w.warehouseCode
45    ORDER BY p.quantityInStock ASC
46    LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

warehouseName	productName	quantityInStock
North	1960 BSA Gold Star DBD34	15
East	1968 Ford Mustang	68
West	1928 Ford Phaeton Deluxe	136
North	1997 BMW F650 ST	178
South	Pont Yacht	414
West	1911 Ford Town Car	540
West	1928 Mercedes-Benz SSK	548
North	F/A 18 Hornet 1/72	551
North	2002 Yamaha YZR M1	600
South	The Mayflower	737

Result Grid | Form Editor | Field Types

ANALYSIS 5: SALES REPRENSENTATIVE PERFORMANCE

This insight enables stakeholders to evaluate individual performance, recognize high achievers, and design incentive programs that drive motivation and productivity. Additionally, it helps identify performance gaps and opportunities for training and development limited to 5 Sales Representative.

```
49      -- 6. SALESREP PERFORMANCE
50      -- ========================================
51  •   SELECT
52          e.firstName,
53          e.lastName,
54          e.jobTitle,
55          SUM(od.quantityOrdered * od.priceEach) AS totalSales
56      FROM employees e
57      JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber
58      JOIN orders o ON c.customerNumber = o.customerNumber
59      JOIN orderdetails od ON o.orderNumber = od.orderNumber
60      GROUP BY e.employeeNumber
61      ORDER BY totalSales DESC
62      LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

firstName	lastName	jobTitle	totalSales
Gerard	Hernandez	Sales Rep	1258577.81
Leslie	Jennings	Sales Rep	1081530.54
Pamela	Castillo	Sales Rep	868220.55
Larry	Bott	Sales Rep	732096.79
Barry	Jones	Sales Rep	704853.91

Result Grid

Form Editor

ANALYSIS 6: PRODUCT HIGHEST DEMAND

Using SQL queries such as SUM, JOIN, GROUP BY, and ORDER BY, I analyzed the seven most in-demand products based on revenue. This analysis provides stakeholders with clear insights into the best-performing products as well as those with lower demand, enabling the team to make more informed strategic decisions aimed at improving underperforming products.

```
74      -- ============================================
75      -- 7. PRODUCT HIGEST DEMAND
76      -- ============================================
77  •   SELECT
78          p.productName,
79          SUM(od.quantityOrdered) AS totalQuantitySold,
80          SUM(od.quantityOrdered * od.priceEach) AS totalRevenue -- This shows the dollar value
81      FROM products p
82      JOIN orderdetails od ON p.productCode = od.productCode
83      GROUP BY p.productName
84      ORDER BY totalQuantitySold DESC
85      LIMIT 7;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

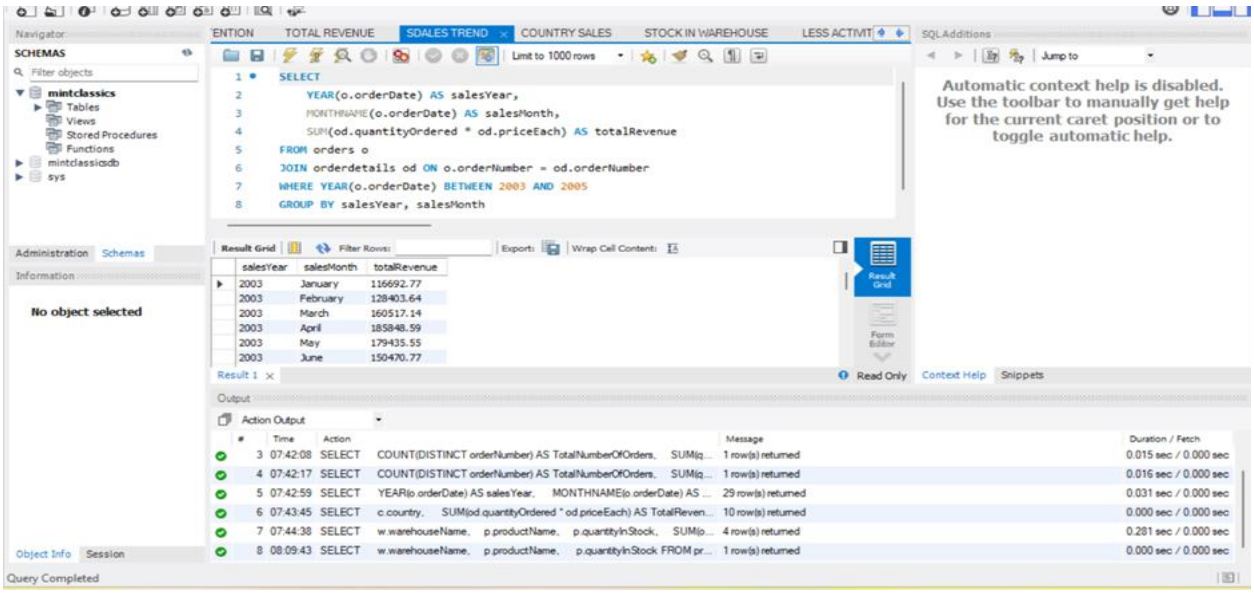
productName	totalQuantitySold	totalRevenue
1992 Ferrari 360 Spider red	1808	276839.98
1937 Lincoln Berline	1111	102563.52
American Airlines: MD-11S	1085	71753.93
1941 Chevrolet Special Deluxe Cabriolet	1076	102537.45
1930 Buick Marquette Phaeton	1074	41599.24
1940s Ford truck	1061	114232.79
1969 Harley Davidson Ultimate Chopper	1057	90157.77

Result Grid

Form Editor

ANALYSIS 7: SALES TREND

SQL (Structured Query Language) was used to efficiently extract and analyze sales data from the Mint Classics dataset for the period 2003 to 2005. By aggregating sales on a yearly basis, this analysis provides a clear view of revenue trends over time, enabling stakeholders to assess business performance and identify growth or decline patterns.



ANALYSIS 8: CREDIT LIMIT

The SQL query was used to analyze and understand customers' credit limits. If credit limits are not properly regulated, they can negatively impact the company's revenue. In this analysis, customers' credit limits were limited to 10 for visualization.

```sql
SELECT
    c.customerName,
    c.creditLimit,
    SUM(od.quantityOrdered * od.priceEach) AS totalOutstandingValue,
    (c.creditLimit - SUM(od.quantityOrdered * od.priceEach)) AS remainingCredit
FROM customers c
JOIN orders o ON c.customerNumber = o.customerNumber
JOIN orderdetails od ON o.orderNumber = od.orderNumber
GROUP BY c.customerName, c.creditLimit
HAVING totalOutstandingValue > (c.creditLimit * 0.8) -- Flag if over 80% used
ORDER BY remainingCredit ASC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

customerName	creditLimit	totalOutstandingValue	remainingCredit
Euro+ Shopping Channel	227600.00	820689.54	-593089.54
Mini Gifts Distributors Ltd.	210500.00	591827.34	-381327.34
Down Under Souveniers, Inc	88000.00	154622.08	-66622.08
The Sharp Gifts Warehouse	77600.00	143536.27	-65936.27
Salzburg Collectables	71700.00	137480.07	-65780.07
Australian Collectors, Co.	117300.00	180585.07	-63285.07
Dragon Souveniers, Ltd.	103800.00	156251.03	-52451.03
Reims Collectables	81100.00	126983.19	-45883.19
Danish Wholesale Imports	83400.00	129085.12	-45685.12
Gifts4AllAges.com	41900.00	84340.32	-42440.32

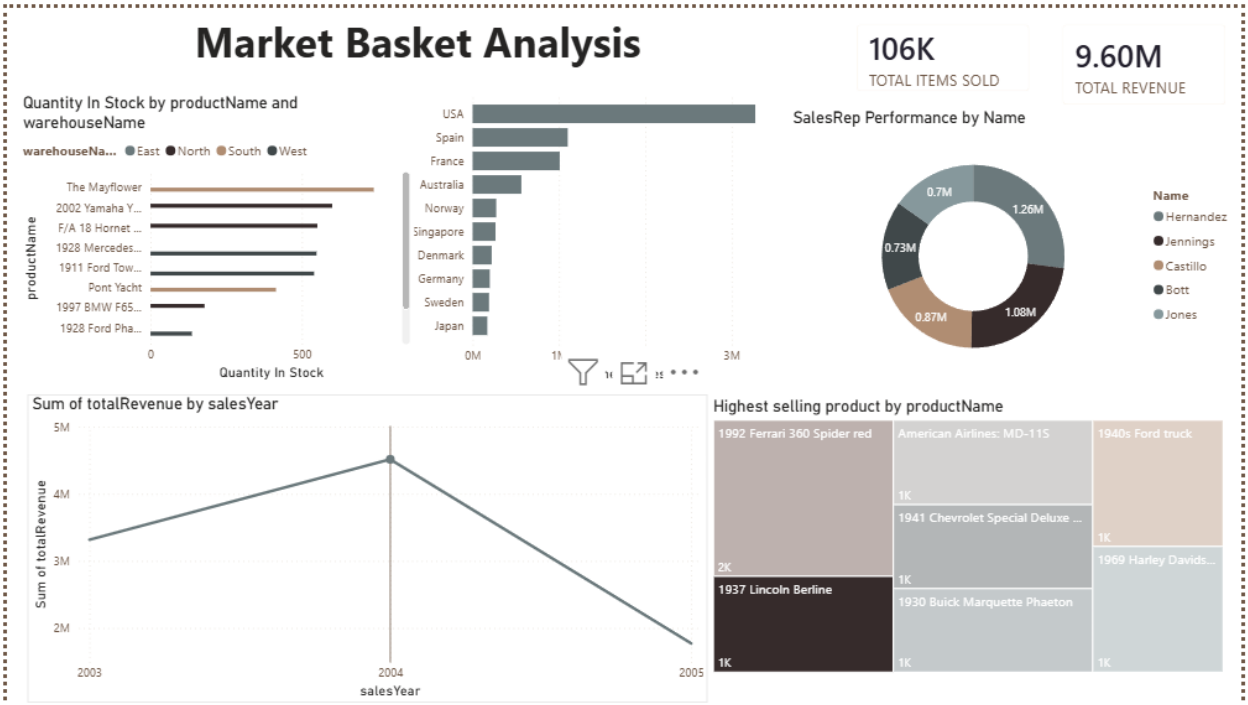
BUSINESS INSIGHTS:

For more detailed insights, the transformed SQL results were exported into a dashboard, as visual representation provides a clearer understanding of complex data. This dashboard offers a simplified yet comprehensive view of the overall project through KPIs and charts, including revenue, warehouse stock levels, highest product demand, sales representative performance and country with the highest sales.

Despite these achievements, **the company is experiencing a downward trend, which is more clearly highlighted in the dashboard** and easier for stakeholders, especially non-technical users, to interpret.

From the visualization, it was observed that sales generally declined around 2005. While SQL queries alone could not fully reveal these insights, the dashboard made it possible to

identify that the company is carrying a high level of uncollected credit risk. **The downward trend in the line chart likely reflects the impact of an unsustainable credit strategy, which is affecting the company's ability to maintain consistent revenue growth.**



RECOMMENDATION

Establish Automated Credit Controls: Implementing a rule-based workflows that automatically flag or block orders for accounts exceeding their credit thresholds. This prevents further exposure until accounts are reviewed.

Continuous Portfolio Monitoring: Move beyond static credit limits by monitoring real-time risk signals, such as payment delays or changes in financial health, to proactively manage your credit portfolio.